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5 / 5 submitted

Q1. Program based on Classes and Objects - Initializaing Data member using Objects directly

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
public class Main{
    public static void main(String[]args){
        Scanner S1=new Scanner(System.in);
        Student s=new Student();

        s.rollNo=S1.nextInt();
        S1.nextLine();
        s.name=S1.nextLine();
        s.marks=S1.nextDouble();

        s.displayDetails();
    }
}
class Student{
    public int rollNo;
    public String name;
    public double marks;

    public void displayDetails(){
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }
}
```

INPUT

```
21
Mohamed Imraan
85.75
```

OUTPUT

```
Roll No: 21
Name: Mohamed Imraan
Marks: 85.75
```

Q2. Classes and Objects-Initialize Object Data Using Setter Methods

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
public class Main{
    public static void main(String[]args){
        Scanner S1=new Scanner(System.in);

        int rollNo=S1.nextInt();
        S1.nextLine();
        String name=S1.nextLine();
        double marks=S1.nextDouble();

        Student s=new Student();

        s.setRollNo(rollNo);
        s.setName(name);
        s.setMarks(marks);

        s.displayDetails();
    }
}
class Student{
    private int rollNo;
    private String name;
    private double marks;

    public void setRollNo(int rollNo){
        this.rollNo=rollNo;
    }
    public void setName(String name){
        this.name=name;
    }
    public void setMarks(double marks){
        this.marks=marks;
    }
    public void displayDetails(){
        System.out.println("Roll No: "+this.rollNo);
        System.out.println("Name: "+this.name);
        System.out.println("Marks: "+this.marks);
    }
}
```

INPUT

20
Imraan
92

OUTPUT

Roll No: 20
Name: Imraan
Marks: 92.0

Q3. Student Details Using Default and Parameterized Constructors

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
public class main{
    public static void main(String[]args){
        Scanner S1=new Scanner(System.in);

        Student s=new Student();
        s.displayDetails();

        int rollNo=S1.nextInt();
        S1.nextLine();
        String name=S1.nextLine();
        double marks=S1.nextDouble();

        Student s2=new Student(rollNo,name,marks);
        s2.displayDetails();
    }
}
class Student{
    private int rollNo;
    private String name;
    private double marks;

    public Student(){
        this.rollNo=101;
        this.name="Rahul";
        this.marks=85.5;
    }
    public Student(int rollNo,String name,double marks){
        this.rollNo=rollNo;
        this.name=name;
        this.marks=marks;
    }
    public void displayDetails(){
        System.out.println("Roll No: "+ this.rollNo);
        System.out.println("Name: "+ this.name);
        System.out.println("Marks: "+ this.marks);
    }
}
```

INPUT

```
21
Imraan
92
```

OUTPUT

```
Roll No: 101
Name: Rahul
Marks: 85.5
Roll No: 21
Name: Imraan
Marks: 92.0
```

Q4. Net Salary Calculation Using Single Inheritance

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
public class Main{
    public static void main(String[]args){
        Scanner S1=new Scanner(System.in);

        int empId=S1.nextInt();
        S1.nextLine();
        String empName=S1.nextLine();
        double basicSalary=S1.nextDouble();

        Salary s=new Salary();
        s.getEmployeeDetails(empId,empName,basicSalary);
        s.calculateSalary();
        s.displayDetails();
    }
}
class Employee{
    int empId;
    String empName;
    double basicSalary;

    public void getEmployeeDetails(int empId,String empName,double empbasicSalary){
        this.empId=empId;
        this.empName=empName;
        this.basicSalary=empbasicSalary;
    }
}
class Salary extends Employee{
    double hra,da,netSalary;

    public void calculateSalary(){
        this.hra=this.basicSalary*0.20;
        this.da=this.basicSalary*0.10;
        this.netSalary=this.basicSalary+this.hra+this.da;
    }
    public void displayDetails(){
        System.out.println("Employee ID: "+this.empId);
        System.out.println("Employee Name: "+this.empName);
        System.out.println("Basic Salary: "+this.basicSalary);
        System.out.println("HRA: "+this.hra);
        System.out.println("DA: "+this.da);
        System.out.println("Net Salary: "+this.netSalary);
    }
}
```

INPUT

101
Imraan
50000

OUTPUT

Employee ID: 101
Employee Name: Imraan
Basic Salary: 50000.0
HRA: 10000.0
DA: 5000.0
Net Salary: 65000.0

Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
public class Main{
    public static void main(String[]args){
        Scanner S1=new Scanner(System.in);

        int empId=S1.nextInt();
        S1.nextLine();
        String empName=S1.nextLine();
        double basicSalary=S1.nextDouble();

        GrossSalary g= new GrossSalary();

        g.getEmployeeDetails(empId,empName);
        g.getBasicSalary(basicSalary);
        g.calculateSalary();
        g.displayDetails();
    }
}
class Employee{
    int empId;
    String empName;

    public void getEmployeeDetails(int empId,String empName){
        this.empId=empId;
        this.empName=empName;
    }
}
class Salary extends Employee{
    double basicSalary;

    public void getBasicSalary(double basicSalary){
        this.basicSalary=basicSalary;
    }
}
class GrossSalary extends Salary{
    double hra, da,grossSalary;

    public void calculateSalary(){
        this.hra=this.basicSalary*0.20;
        this.da=this.basicSalary*0.10;
        this.grossSalary=this.basicSalary+this.hra+this.da;
    }

    public void displayDetails(){
        System.out.println("Employee ID: "+this.empId);
        System.out.println("Employee Name: "+this.empName);
        System.out.println("Basic Salary: "+this.basicSalary);
        System.out.println("HRA: "+this.hra);
        System.out.println("DA: "+this.da);
        System.out.println("Gross Salary: "+this.grossSalary);
    }
}
```

INPUT

```
101
Kaif
50000
```

OUTPUT

```
Employee ID: 101
Employee Name: Kaif
Basic Salary: 50000.0
HRA: 10000.0
```